



Masoud Aghighi

**HVAC Technician | HVAC Controls | FDD
& AI-Assisted Diagnostics**

Contact



masoudaghighi20@gmail.com



(+1)2369711260



BC, Canada

Skills

**HVAC Troubleshooting &
Diagnostics**

**HVAC Installation &
Maintenance**

**Electrical & Mechanical
Troubleshooting**

Furnaces Diagnostics

**HRV / ERV & Ventilation
installation**

HVAC Controls & Thermostats

**Water-Cooled Chillers, Rooftop
Units economizer &
Optimization**

**Airflow Testing &
Troubleshooting & Optimization**

**Ductwork and Sheet Metal
Fabrication & Assembly**

Python & FDD – Developing

**AI-Assisted HVAC Diagnostics –
Developing**

Profile Summary

HVAC Technician with hands-on experience in installation, maintenance, troubleshooting, and repair of residential, commercial, and industrial HVAC systems.

Experienced with furnaces, heat pumps, air conditioning, HRV/ERV, ventilation, and RTUs.

Strong background in electrical, mechanical, refrigeration, airflow, and HVAC controls troubleshooting.

Previous experience in electronics, smart thermostats, HVAC controls, firmware, and testing.

Currently developing skills in BAS, FDD, Python, and AI-assisted HVAC diagnostics.

Focused on combining practical HVAC knowledge with automation and intelligent diagnostic technologies.

Education

Associate in Electronics Technician- Electronics

Branch: IAU University
Ashtian, Markazi, Iran
2005

Work Experience

HVAC Technician

Steam Dry Canada
Vancouver, BC, Canada
2025 - 2026

Tasks and Achievements

Diagnose and repair electrical, mechanical, airflow, and performance issues in furnaces and heat pumps.

Perform preventive maintenance, refrigerant diagnostics, coil cleaning, parts replacement, and system optimization.

Perform complete HVAC system cleaning and sanitizing to maintain system performance and reliability.

Communicate technical findings clearly with customers and maintain professional, organized job sites.

Certificates

SkilledTradesBC ITA – HVAC Trade Qualification Pathway(RedSeal)

2025

WHMIS (GHS) – 2025

2023

Specialized Reverse Engineering Course – 2023

2025

BC Class 4 Driver's License – Clean Driving Record

2025

Valid Class 5 license, with a clean driving record (BC)

2023

Professional Technical Skills in Public Electronics Course Certification (438 hours), 2003

2023

First Aid & CPR/AED

Social Network



masoud-aghighi322



masoud678/HVAC-fault-detection

HVAC Assistant

Vortex HVAC Solutions
Vancouver, BC, Canada

2024 - 2025

Tasks and Achievements

Assisted with installation, maintenance, troubleshooting, and repair of HVAC and ventilation systems.

Diagnosed faults involving motors, fans, control boards, capacitors, fuses, and other electrical components.

Performed coil cleaning, refrigerant diagnostics, system performance checks, and sheet metal fabrication and installation.

Installed HRV/ERV systems, ventilation fans, sheet metal components, fresh-air intake, and exhaust piping.

Electromechanical Technician

Padideh Pars Company
Tehran, Iran

2017 - 2022

Tasks and Achievements

Maintained and repaired residential, commercial, and industrial HVAC equipment, including chillers, heaters, and RTUs.

Troubleshoot three-phase motors, electrical systems, pumps, bearings, couplings, and cooling equipment.

Read technical drawings and wiring diagrams and performed HVAC piping, soldering, welding, and hydronic heating work.

Electrical Technician

Marsus Smart Home Co.
Iran

2012 - 2017

Tasks and Achievements

Designed, assembled, and tested analog and digital thermostats and HVAC control systems.

Developed and calibrated smart thermostat firmware and supported troubleshooting and system upgrades.

Worked with engineering teams on thermostat prototyping, testing, manufacturing, and quality control.

Projects

HVAC FDD / AI PROJECT

Developing a Python-based HVAC FDD system for sensor validation, temperature deviation analysis, cooling fault detection, and fault severity classification, with future development toward AI-assisted diagnostics.

Link: <https://github.com/masoud678/HVAC-fault-detection>